The left followed by the right distributive laws yield $(1+1)(a+b) = (1+1)a+(1+1)b = a+a+b+b$. The right followed by the left distributive laws yield $(1+1)(a+b) = 1(a+b)+1(a+b) = a+b+a+b$. Thus $a+a+b+b = a+b+a+b$ and by cancellation in the additive group, we obtain $a+b = b+a$.

55. Let $a, b \in R$ where R is a Boolean ring. We have $a+b = (a+b)^2 = a^2+ab+ba+b^2 = a+ab+ba+b$. Thus in a Boolean ring, $ab = -ba$. Taking $b = a$, we see that $aa = -aa$, so $a = -a$. Thus every element is its own additive inverse, so $-ba = ba$. Combining our equations $ab = -ba$ and $-ba = ba$, we obtain $ab = ba$, showing that R is commutative.

56. a.

$+$	$\varnothing$	$\{a\}$	$\{b\}$	S
$\varnothing$	$\varnothing$	$\{a\}$	$\{b\}$	S
$\{a\}$	$\{a\}$	$\varnothing$	S	$\{b\}$
$\{b\}$	$\{b\}$	S	$\varnothing$	$\{a\}$
S	S	$\{b\}$	$\{a\}$	$\varnothing$

$\cdot$	$\varnothing$	S	$\{a\}$	$\{b\}$
$\varnothing$	$\varnothing$	$\varnothing$	$\varnothing$	$\varnothing$
S	$\varnothing$	S	$\{a\}$	$\{b\}$
$\{a\}$	$\varnothing$	$\{a\}$	$\{a\}$	$\varnothing$
$\{b\}$	$\varnothing$	$\{b\}$	$\varnothing$	$\{b\}$

b. Let $A, B \in \mathcal{P}(S)$. Then $A+B = (A \cup B) - (A \cap B) = (B \cup A) - (B \cap A) = B+A$, so addition is commutative.

We check associativity of addition; it is easiest to think in terms of the elements in $(A+B)+C$ and the elements in $A+(B+C)$. By definition, the sum of two sets contains the elements in precisely one of the sets. Thus $A+B$ consists of the elements that are in either one of the sets A or B, but not in the other. Therefore $(A+B)+C$ consists of the elements that are in precisely one of the three sets A, B, C. Clearly $A+(B+C)$ yields this same set, so addition is associative.

The empty set $\varnothing$ acts a additive identity, for $A+\varnothing = (A \cup \varnothing) - (A \cap \varnothing) = A - \varnothing = A$ for all $A \in \mathcal{P}(S)$.

For $A \in \mathcal{P}(S)$, we have $A+A = (A \cup A) - (A \cap A) = A - A = \varnothing$, so each element of $\mathcal{P}(S)$ is its own additive inverse. This shows that $\langle \mathcal{P}(S), + \rangle$ is an abelian group.

For associativity of multiplication, we see that $(A \cdot B) \cdot C = (A \cap B) \cap C = A \cap (B \cap C) = A \cdot (B \cdot C)$.

For the left distributive law, we again think in terms of the elements in the sets. The set $A \cdot (B+C) = A \cap (B+C)$ consists of all elements of A that are in precisely one of the two sets B, C. This set thus contains all the elements in $A \cap B$ or in $A \cap C$, but not in both sets. This is precisely the set $(A \cap B) + (A \cap C) = (A \cdot B) + (A \cdot C)$. The right distributive law can be demonstrated by a similar argument.

We have shown that $\langle \mathcal{P}(S), +, \cdot \rangle$ is a ring. Because $A \cdot A = A \cap A = A$, we see from the definition in Exercise 55 that it is a Boolean ring.

19. Integral Domains

1. We rewrite the equation as $x(x-3)(x+1) = 0$, and simply try all the elements, -5 -4, -3 -2, -1, 0, 1, 2, 3, 4, 5, 6 of $\mathbb{Z}_{12}$, obtaining the solutions 0, 3, 5, 8, 9, and 11.

2. The solution in $\mathbb{Z}_7$ is 3 and the solution in $\mathbb{Z}_{23}$ is 16.

3. Trying all possibilities -2, -1, 0, 1, 2, and 3, we find no solutions.

4. Trying all possibilities -2, -1, 0,1, 2, and 3, we find $x = 2$ as the only solution.

5. 0 **6.** 0 **7.** 0 **8.** 3 **9.** 12 **10.** 30

11. $(a+b)^4 = a^4 + 4 \cdot a^3b + 6 \cdot a^2b^2 + 4 \cdot ab^3 + b^4 = a^4 + 2 \cdot a^2b^2 + b^4$

12. $(a+b)^9 = [(a+b)^3]^3 = [a^3 + 3 \cdot a^2b + 3 \cdot ab^2 + b^3]^3 = (a^3 + b^3)^3 = a^9 + 3 \cdot a^6b^3 + 3 \cdot a^3b^6 + b^9 = a^9 + b^9$.

13. $(a+b)^6 = [(a+b)^3]^2 = [a^3 + 3 \cdot a^2b + 3 \cdot ab^2 + b^3]^2 = (a^3 + b^3)^2 = a^6 + 2 \cdot a^3b^3 + b^6$.

14. We have $\begin{bmatrix} 2 & -1 \\ 2 & -1 \end{bmatrix} \begin{bmatrix} 1 & 2 \\ 2 & 4 \end{bmatrix} = \begin{bmatrix} 0 & 0 \\ 0 & 0 \end{bmatrix}$.

15. The definition is incorrect. We must state that $a \neq 0$ and $b \neq 0$.

If, in a ring R, nonzero elements a and b are such that $ab = 0$, then a and b are **divisors of zero**.

16. The definition is incorrect; n must be minimal in $\mathbb{Z}^+$.

If for some $n \in \mathbb{Z}^+, n \cdot a = 0$ for all a in a ring R, the smallest such n is the **characteristic** of R. If no such n exists, then 0 is the **characteristic** of R.

17. F T F F T T F T F F

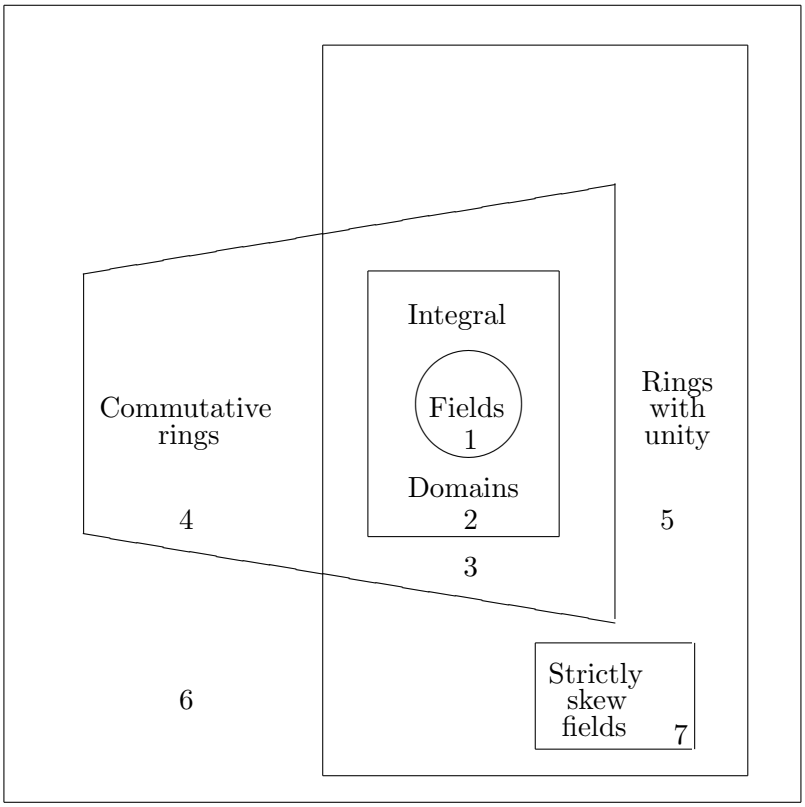
18. Specifying the regions by their number, we have as examples:

1. $\mathbb{Q}$ 2. $\mathbb{Z}$ 3. $\mathbb{Z}_4$ 4. $2\mathbb{Z}$ 5. $M_2(\mathbb{R})$ and

6. Upper-triangular matrices with integer entries and all zeros on the main diagonal.

19. (See the answer in the text.)

20. We need add only one region, No. 7, to the existing figure. Unfortunately, we can't draw big circles, so we had to use quadrilaterals which are a bit confusing. For example, the commutative ring quadrilateral extends into the high rectangle, and includes regions numbered 1, 2, 3, and 4. This would be apparent if we could have made it a large circle. We suggest that you draw circles, as in the text. Just add another region 7 inside the existing region 5 of the text, and label it "Strictly skew fields".



21. Rewriting $ab = ac$ as $a(b - c) = 0$, deduce $b = c$ from $a \neq 0$ and the absence of zero divisors.

22. If $a \neq 0$ is in the finite integral domain D, use the cancellation law and cardinality to deduce that the map of D into D where x is mapped into ax is one-to-one and onto D.

23. If $a^2 = a$, then $a^2 - a = a(a-1) = 0$. If $a \neq 0$, then a^{-1} exists in R and we have $a-1 = (a^{-1}a)(a-1) = a^{-1}[a(a-1)] = a^{-1}0 = 0$, so $a - 1 = 0$ and $a = 1$. Thus 0 and 1 are the only two idempotent elements in a division ring.

24. Exercise 49(a) in Section 18 showed that an intersection of subrings of a ring R is again a subring of R. Thus an intersection of subdomains D_i for $i \in I$ of an integral domain D is at least a ring. The preceding exercise shows that unity in an integral domain can be characterized as the nonzero idempotent. This shows that the unity in each D_i must be the unity 1 in D, so 1 is in the intersection of the D_i. Of course multiplication is commutative in the intersection because it is commutative in D and the operation is induced. Finally, if $ab = 0$ in the intersection, then $ab = 0$ in D so either $a = 0$ or $b = 0$, that is, the intersection has no divisors of zero, and is a subdomain of D.

25. Because R has no divisors of zero, multiplicative cancellation of nonzero elements is valid. The construction in the proof of Theorem 18.11 is valid and shows that each nonzero $a \in R$ has a right inverse, say a_i. A similar construction where the elements of R are all multiplied on the *right* by a shows that a has a left inverse, say a_j. By associativity of multiplication, we have $a_j = a_j(aa_i) = (a_ja)a_i = a_i$. Thus every nonzero $a \in R$ is a unit, so R is a division ring.

26. **a.** Let $a \neq 0$. We wish to show that a is not a divisor of zero. Let b be the unique element such that $aba = a$. Suppose $ac = 0$ or $ca = 0$. Then $a(b+c)a = aba + aca = a + 0 = a$. By uniqueness $b + c = b$ so $c = 0$.

b. From $aba = a$, we know that $b \neq 0$ also. Multiplying on the left by b, we obtain $baba = ba$. Because R has no divisors of zero by part **a**, multiplicative cancellation is valid and we see that $bab = b$.

c. We claim that ab is unity for nonzero a and b given in the statement of the exercise. Let $c \in R$. From $aba = a$, we see that $ca = caba$. Cancelling a, we obtain $c = c(ab)$. From part **b**, we have $bc = babc$, and cancelling b yields $c = (ab)c$. Thus ab satisfies $(ab)c = c(ab)$ for all $c \in R$, so ab is unity.

d. Let a be a nonzero element of the ring. By part **a**, $aba = a$. By part **c** , $ab = 1$ so b is a right inverse of a. Because the elements a and b behave in a symmetric fashion by part **b**, an argument symmetric to that in part **c**, starting with the equation $ac = abac$, shows that $ba = 1$ also. Thus b is also a left inverse of a, so a is a unit. This shows that R is a division ring.

27. By Exercise 23, we see that the unity in an integral domain can be characterized as the unique nonzero idempotent. The unity element in D must then also be the unity in every subdomain. Recall that the characteristic of a ring with unity is the minimum $n \in \mathbb{Z}^+$ such that $n \cdot 1 = 0$, if such an n exists, and is 0 otherwise. Because unity is the same in the subdomain, this computation will lead to the same result there as in the original domain.

28. Let $R = \{n \cdot 1 \mid n \in \mathbb{Z}\}$. We have $n \cdot 1 + m \cdot 1 = (n+m) \cdot 1$ so R is closed under addition. Taking $n = 0$, we see that $0 \in R$. Because the inverse of $n \cdot 1$ is $(-n) \cdot 1$, we see that R contains all additive inverses of elements, so $\langle R, + \rangle$ is an abelian group. The distributive laws show that $(n \cdot 1)(m \cdot 1) = (nm) \cdot 1$, so R is closed under multiplication. Because $1 \cdot 1 = 1$, we see that $1 \in R$. Thus R is a commutative ring with unity. Because a product $ab = 0$ in R can also be viewed as a product in D, we see that R also has no divisors of zero. Thus R is a subdomain of D.

29. Suppose the characteristic is mn for $m > 1$ and $n > 1$. Following the hint, the distributive laws show that $(m \cdot 1)(n \cdot 1) = (nm) \cdot 1 = 0$. Because we are in an integral domain, we must have either $m \cdot 1 = 0$ or $n \cdot 1 = 0$. But if $m \cdot 1 = 0$ then Theorem 19.15 shows that the characteristic of D is at most m. If $n \cdot 1 = 0$, the chracteristic of D is at most n. Thus the characteristic can't be a composite positive integer, so it must either be 0 or a prime p.

30. **a.** From group theory, we know that S is an abelian group under addition. We check the associativity of multiplication, using the facts that, for all $m, n \in \mathbb{Z}$ and $r, s \in R$, we have $n \cdot (m \cdot r) = (nm) \cdot r, n \cdot (r+s) = n \cdot r + n \cdot s, r(n \cdot s) = n \cdot (rs)$, and $(n \cdot r)s = n \cdot (rs)$, which all follow from commutativity of addition and the distributive laws in R. We have, for $r, s, t \in R$ and $k, m, n \in \mathbb{Z}$,

$$(r,k)[(s,m)(t,n)] = (r,k)(st + m \cdot t + n \cdot s, mn)$$
$$= (r(st + m \cdot t + n \cdot s) + k \cdot (st + m \cdot t + n \cdot s) + mn \cdot r, kmn)$$
$$= (rst + k \cdot st + m \cdot rt + n \cdot rs + km \cdot t + kn \cdot s + mn \cdot r, kmn)$$

and

$$[(r,k)(s,m)](t,n) = (rs + k \cdot s + m \cdot r, km)(t,n)$$
$$= ((rs + k \cdot s + m \cdot r)t + km \cdot t + n \cdot (rs + k \cdot s + m \cdot r), kmn)$$
$$= (rst + k \cdot st + m \cdot rt + n \cdot rs + km \cdot t + kn \cdot s + mn \cdot r, kmn).$$

Thus multiplication is associative. For the left distributive law, we obtain

$$\begin{aligned}(r,k)[(s,m) + (t,n)] &= (r,k)(s+t, m+n) \\ &= (r(s+t) + k \cdot (s+t) + (m+n) \cdot r, k(m+n) \\ &= (rs + k \cdot s + m \cdot r, km) + (rt + k \cdot t + n \cdot r, kn) \\ &= (r,k)(s,m) + (r,k)(t,n).\end{aligned}$$